REGIME-SWITCHING RISK MODELS FOR ETF PORTFOLIOS: DO MARKET-STATE SHIFTS IMPROVE TAIL-RISK FORECASTING?

by

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ABSTRACT

Stock–bond correlation turned positive in 2022, which raises the question of whether tail-risk forecasts improve when the correlation is allowed to switch with a market regime. I forecast one-day value-at-risk and expected shortfall for an equal-weight portfolio of SPY, IEF and GLD and compare two models with the same GARCH margins: constant SPY–IEF correlation, and a correlation that switches with a two-state hidden Markov regime fitted to SPY. Forecasts are scored with the FZ0 loss. The comparison, its 5% tail level, its decision rule, a smallest effect of interest and a simulated power study were fixed before the 2020–2025 hold-out was used. On the 1,508 hold-out days the switching model's loss was lower by 0.0013 (95% interval −0.0037 to +0.0012; one-sided $p = 0.158$). The test did not reject. It could detect a difference of 0.0031 with 80% probability, and an improvement equal to the smallest effect of interest, 0.0020, would have been detected with probability about 48%, so the data are consistent with a gain and with none. Both models breached the 5% VaR on about 6.8% to 6.9% of hold-out days, and breach-day losses exceeded the forecast expected shortfall; a post hoc EWMA baseline breached on 5.2% of days, with a loss 0.0063 above that of constant correlation (standard error 0.0440). On the development data, regime-specific volatility lowered the loss against a pooled normal in three of four ETF samples, but GARCH margins did better still, and correlation steps were small against their detectable differences. The thesis reports what a sample of this length can and cannot detect, and does not claim that regime-switching correlation is absent or present.

KEYWORDS: Value-at-Risk; Expected Shortfall; Regime Switching; Forecast Evaluation; Statistical Power

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LIST OF ABBREVIATIONS AND NOTATION

Abbreviations

CCC	constant conditional correlation model
A, B, D	static covariance models: A is a pooled normal, B has regime-specific volatilities, D has regime-specific volatilities and correlations (Chapter 4)
DCC	dynamic conditional correlation model
DM	Diebold-Mariano test
ES	expected shortfall
ETF	exchange-traded fund
FZ0	the Fissler-Ziegel-type joint loss for VaR and ES used here (Chapter 5)
GARCH	generalised autoregressive conditional heteroskedasticity
GW	Giacomini-White test of conditional predictive ability
HAC	heteroskedasticity- and autocorrelation-consistent (variance estimator)
HMM	hidden Markov model
MDE	minimum detectable difference
PSD	positive semi-definite
QML	quasi-maximum likelihood
RSC	regime-switching correlation model (all correlations switch)
RSC_P	regime-switching correlation model in which only the SPY–IEF correlation switches (the primary model)
SPY, IEF, GLD	the three funds of the portfolio: US equities, US Treasuries (7–10 years), gold
SESOI	smallest effect of interest
VaR	value-at-risk

Notation (full definitions at first use in Chapters 4 and 5)

t	forecast day; the forecast for day t uses returns through day $t-1$
y_t	portfolio simple return on day t
r_t	vector of the three asset returns (SPY, IEF, GLD)
w	portfolio weights, $(1/3, 1/3, 1/3)$, constant
α	tail level of the VaR and ES forecasts (5%, secondary 2.5%)
VaR_t, ES_t	forecast value-at-risk and expected shortfall of y_t (negative numbers)
S_t	hidden regime, 1 calm, 2 stress
d_t	loss difference, richer minus baseline: $d_t = L_t^{\mathrm{rich}} - L_t^{\mathrm{base}}$; negative favours the richer model
$\bar{d}$, $\widehat{\mathrm{SE}}$	mean loss difference and its HAC standard error
T	number of forecast days in an evaluation sample

1 INTRODUCTION

A risk manager who reports a daily value-at-risk (VaR) and expected shortfall (ES) for a portfolio of stocks, Treasuries and gold has to say how those assets move together in a bad week. The correlation between stock and bond returns has no fixed sign. It was about 60% in the mid-1990s and about -60% in the early 2000s (Baele et al., 2010), and it switched from positive to negative around 2001 (Campbell et al., 2020). It turned positive again in 2022: the correlation of daily SPY and IEF returns was -0.46 over 2010 to 2019 and $+0.19$ in 2022, and it stayed near zero in 2023 ($+0.04$) and 2024 ($+0.07$) (own calculation from the data of Chapter 3, descriptive only). A forecast that holds the correlation constant would be too optimistic in the tail on exactly the days when the bond holding stops offsetting equity losses. The risk measure matters to practitioners beyond internal reporting: the Basel Committee's market-risk standard sets capital for internal models on expected shortfall at the 97.5% level, a 2.5% tail (Basel Committee on Banking Supervision, 2019), which is the secondary tail level used here. In the hold-out of this thesis, 2020 to 2025, the correlation between the US equity fund SPY and the Treasury fund IEF over the previous 63 trading days was positive on 44.2% of days (Chapter 7).

This thesis asks whether letting the SPY–IEF correlation switch with a market regime improves one-day-ahead VaR and ES forecasts of a fixed portfolio, an equal-weight holding of SPY, IEF and the gold fund GLD. The confirmatory comparison was fixed before I used the test sample: a model in which the SPY–IEF correlation depends on the state of a two-state hidden Markov model fitted to SPY returns, against a model with constant correlation and the same GARCH margins. A wider ladder of models, run on development data only, shows where in the chain of models regime information helps; those results are descriptive.

The economic reasoning comes from the stock–bond literature. Campbell et al. (2020) tie the sign of the correlation to a slowly changing macroeconomic state, while Baele et al. (2010) find that macroeconomic fundamentals explain little and that market conditions such as liquidity

and the variance premium matter more. A two-state volatility regime is a market-based signal, so the test speaks to the second reading and not the first (Chapter 8). The model also learns the stress-day correlation from the five years before each forecast, and before 2022 that correlation was negative or low (Table 6.7). The question is therefore whether the model improves forecasts, not whether it reproduces the 2022 positive correlation.

The literature leaves the question open (Chapter 2). Regime switching improves volatility-based risk forecasts (Ardia et al., 2018), regime-switching correlation has been proposed (Pelletier, 2006) and tested for densities and VaR mainly on stocks (Paolella et al., 2019), and univariate and multivariate models have been compared for expected shortfall without regime switching (Fortin et al., 2023). I did not find a study of regime-switching correlation that scores VaR and ES jointly and reports what the test could and could not detect.

Forecasts are scored with the FZ0 loss of Patton et al. (2019), which scores VaR and ES jointly, and models are compared with a Diebold–Mariano test with a heteroskedasticity-and-autocorrelation-consistent standard error. The sample is split into development days, on which I explored and which validate nothing, and a hold-out of 1,508 days, 2020 to 2025, used once.

Switching the SPY–IEF correlation lowered the average loss by 0.0013, with a 95% interval from -0.0037 to $+0.0012$, and the one-sided test did not reject ($p = 0.158$). A loss difference of that size is roughly the penalty for scaling both the VaR and ES forecasts by a factor between 2 and 3 per cent from the truth (Section 5.7). The smallest difference the test could detect with 80% probability was 0.0031, and a true improvement equal to the smallest effect of interest, 0.0020, would have been detected with probability about 48%. The data are consistent with a gain and with none. The result is a measure of how precisely this sample can speak, and I do not read it as evidence that the effect is absent.

The second finding is the more useful one for a practitioner. Both models breached the 5% VaR on about 6.8% to 6.9% of hold-out days, so the comparison is between two forecasts that fell short in the same way. A RiskMetrics EWMA of the portfolio return, scored post hoc on the

same days, breached on 5.2% and had a loss that cannot be told apart from theirs (Section 7.4). On the development days, regime-specific volatility lowered the loss against a pooled normal in three of four ETF samples, GARCH margins did better still (significant after adjustment in one sample), and the correlation steps were small beside the differences they could detect.

The contribution is an estimate with an interval and a stated detection limit for a question the literature leaves open, not a better model. Chapter 2 reviews the literature, Chapters 3 to 5 set out the data, models and evaluation, Chapters 6 and 7 report the results, and Chapters 8 and 9 discuss and conclude.

2 LITERATURE REVIEW

This chapter positions the thesis against the work closest to it. It follows the argument of the thesis: what regime switching adds to risk forecasts; how much of that gain survives a strong baseline; how correlation has been modelled and where regime-switching correlation has been evaluated; what is known about the stock–bond correlation itself; how forecasts are evaluated and compared; and how null results are best reported. Where a claim depends on a source I could read only as a working paper or an abstract, I say so, and the references mark working-paper versions.

2.1 Regime-switching volatility and risk forecasting

Markov switching goes back to Hamilton (1989), and regime shifts matter for asset allocation (Ang and Bekaert, 2002), where regime-switching correlations and volatilities rise in bad times. For risk forecasts, Ardia et al. (2018) report that Markov-switching GARCH models give more accurate VaR, ES and left-tail forecasts than single-regime counterparts, especially for stock returns, and that the improvement is larger when switching is added to simple specifications such as a normal GARCH. They score with the loss of Patton et al. (2019), which I also use. Their object is the volatility of a single return series. Whether letting the dependence between assets switch adds anything is a different question, and it is the one this thesis asks.

2.2 Gains relative to the strength of the baseline

The size of a regime gain depends on what it is measured against. Ardia et al. (2018) find the gain largest over simple baselines. In the multivariate literature, Fortin et al. (2023) describe earlier studies of univariate against multivariate volatility models for portfolio VaR as giving mixed results; I take this characterisation from their text and not from the studies. In their own evaluation of weekly portfolio forecasts (the 2021 working paper), Fortin et al. report that static multivariate models tend to be dominated by univariate and dynamic multivariate ones, and that dynamic multivariate models are mostly not significantly different from a univariate skewed-t GARCH-type model. A single-regime GARCH model is therefore a strong baseline, and the confirmatory comparison of Chapter 4 gives both models the same GARCH margins, so that a gain cannot be credited to a weak benchmark.

2.3 Correlation modelling and regime-switching correlation

Correlation can be held constant (Bollerslev, 1990), driven by a GARCH-type recursion (Engle, 2002), or switched between states. Pelletier (2006) proposes the last: a Markov chain selects one of several correlation matrices, each constant within its regime, combined with separate models for the standard deviations. A restricted version scales all correlations by a single regime factor, and the two-step estimation separates the correlation parameters from the volatility ones. In the version I read, the model is applied to four exchange rates and compared with a DCC model by in-sample likelihood and by a non-nested test; the regime model fits better in sample and gives smoother correlations. That version has no out-of-sample forecast comparison and no VaR or ES evaluation. It also notes that autocorrelations of the cross-products of standardised residuals have little power to reveal dynamic correlation. My model RSC is of the same family, but it differs from his in two ways: the regime signal is a hidden Markov model fitted to SPY returns and not a chain estimated jointly with the correlations, and the confirmatory version switches a single pair of correlations and not the whole matrix.

The closest design that I have read is the working-paper version of Paolella et al. (2019): GARCH(1,1) margins that do not depend on the regime, regime-specific correlation matrices, and a forecast that mixes the regime densities with predicted regime probabilities. In 29 Dow Jones stocks they report better density forecasts from regime-switching correlation than from constant correlation. For value-at-risk they report that accuracy is largely determined by the heavy tails of the return distribution and not by the correlation dynamics, and that regime switching does not generally improve the backtest outcome. They score with log scores and coverage tests and not with a joint VaR and ES loss, and they use mainly 29 Dow Jones stocks, with some indices and exchange rates, and no stock–bond portfolio.

The evaluation that is closest in method is Fortin et al. (2023), which compares three univariate with eight multivariate copula models, several with dynamic correlation, for weekly Value-at-Risk and expected shortfall of portfolios of equity factors and of stocks. I describe the 2021 working-paper version. It uses 1,905 weekly out-of-sample forecasts for 1983–2019, tail levels of 1% and 2.5%, the joint loss of Patton et al. (2019), the conditional predictive ability test of Giacomini and White (2006) and the model confidence set of Hansen et al. (2011). It reports that in most cases the differences are not significant, and concludes that the differences are mostly not significant. The design leaves two things open that I take up. It does not use regime-switching correlation, and, in the versions I read, it reports average losses, p-values and sets without an interval for the loss difference and without a power calculation, so a non-significant result is not shown to be small.

2.4 The sign and drivers of the stock–bond correlation

The stock–bond correlation has changed sign. Baele et al. (2010) document realised correlations of about 60% in the mid-1990s and about -60% in the early 2000s, and Campbell et al. (2020) report that the correlation of bond and stock returns was positive on average before the second quarter of 2001 and negative afterwards. The two studies disagree about the drivers. Campbell et al. (2020) link the switch to a change in the sign of the correlation between inflation and

the output gap, dated to 2001 with a break test on macroeconomic data, and show in a habit-formation model that time-varying risk premia amplify the change. Baele et al. (2010) study a dynamic factor model with ten state variables and find that macroeconomic fundamentals explain little of the stock–bond correlation, that liquidity variables matter more, and that a model with constant exposures cannot generate negative correlations; they obtain them when the exposures depend on the variance premium. Neither study forecasts risk, and both end before 2012.

Two points follow. The correlation has changed sign over multi-year spells, which motivates letting it switch. But whether the state behind the change is a slow macroeconomic one or a faster market-based one is not settled: the evidence in one study points to the macroeconomy and in the other away from it. This bears on the regime signal. Connolly et al. (2005) relate the stock–bond return correlation to stock-market uncertainty, measured by implied volatility, which fits the market-based reading; I know that study at the level of its abstract. A two-state model of SPY volatility ties the correlation regime to equity-market stress in the same way. Forbes and Rigobon (2002) show that heteroskedasticity biases correlation-based tests of contagion, a reminder that correlation estimated in volatile periods needs care. I do not test either reading.

2.5 Forecast evaluation and tests of predictive ability

Joint elicitability of VaR and ES (Fissler and Ziegel, 2016) underlies the loss of Patton et al. (2019), whose FZ0 form gives loss differences that are homogeneous of degree zero when forecasts are negative. Coverage is tested by Kupiec (1995), and Christoffersen (1998) tests whether the indicator of a breach is independent and identically distributed with the right probability against a first-order Markov alternative; he shows in a simulation that the independence test needs large samples. These tests describe one forecast at a time. Equal predictive ability is tested by Diebold and Mariano (1995), who allow general loss functions and correlated forecast errors, and West (1996) extends inference to non-nested and nonlinear models whose parameters are estimated. Giacomini and White (2006) take the forecasting method, including its estimation window, as

the object of evaluation and allow models of either kind with a finite rolling window, which is the setting of my test (McCracken, 2020 shows that the unconditional statistic can diverge for a fixed finite window, and that the rolling-window case remains valid). For nested linear models under squared-error loss, Clark and McCracken (2001) and Clark and West (2007) show that normal critical values are inaccurate and propose an adjustment. The primary pair is nested in one correlation entry, but my loss is not squared error and the models are not linear, so I check size in simulation instead of relying on those results (Chapter 5). Superior predictive ability across several candidates is tested by Hansen (2005).

2.6 Negative results, power and pre-registration

Hoenig and Heisey (2001) argue that power calculations made after the experiment to interpret a non-significant result are fundamentally flawed, which is why Chapter 5 reports power only for stated effect sizes. Altman and Bland (1995) make the point that absence of evidence is not evidence of absence, Ioannidis et al. (2017) report that the median power across 159 economics literatures is 18% or less, and Abadie (2020) argues that non-significant results can be highly informative in large-sample settings and should be reported visibly. Lakens (2017) proposes equivalence tests with bounds set by the smallest effect of interest, which is the idea behind the bound in outcome O3, although I do not claim equivalence. Nosek et al. (2018) describe pre-registration as separating prediction from postdiction. I know these six sources at the level of abstracts.

2.7 What this thesis adds

I did not find a study that does all of the following at once: separates the treatment of volatility, correlation and the mean under one regime signal; evaluates VaR and ES forecasts of a fixed portfolio with a joint loss and a test whose estimand, sidedness and decision rule were fixed before the test sample was used; and reports minimum detectable differences and simulated power next to every result. This is a statement about what I read, not a claim that no such

study exists. The sources above leave the confirmatory question open: whether letting the stock–bond correlation switch with an equity-volatility regime improves tail-risk forecasts over constant correlation, with the same GARCH margins in both models, in a period that contains the 2022 reversal. Chapters 5 and 7 give the design and the answer.

3 DATA AND SAMPLE DESIGN

This chapter describes the data, the portfolio and the division of the sample into a development period, a dry-run period and a hold-out period. Many specifications were examined on the development period before one was chosen, so the hold-out is clean only in the sense that nothing was fitted on it.

3.1 Data

The main sample is three exchange-traded funds: SPY (US large-cap equities), IEF (US Treasuries with seven to ten years to maturity) and GLD (gold). I downloaded daily adjusted closing prices from Yahoo Finance through the `yfinance` package on 26 September 2026; the file is stored with its SHA-256 hash in `data/raw/MANIFEST.txt`. The estimation code truncates the series at 31 December 2025 before computing anything.

I compute simple daily returns, $r_{i,t} = P_{i,t}/P_{i,t-1} - 1$, from the adjusted close. Dates on which any of the three prices is missing are dropped and returns are computed between consecutive available dates, with no forward filling. Through 31 December 2019 the file yields 3,805 prices and 3,804 returns, the first on 19 November 2004.

A second download of the same three tickers, truncated at 2019, has the same 3,804 dates, and its daily returns differ from the first by at most 2.5×10^{-6} (portfolio returns by at most 9.3×10^{-7}). This shows that the download is reproducible. It is not a second source: every ETF series in this thesis comes from one vendor.

3.2 The portfolio

The portfolio holds the three funds with constant weights $w = (1/3, 1/3, 1/3)$. The portfolio return is $y_t = w' r_t$. Holding the weights constant means that the portfolio is rebalanced to equal weights every day. I do not model the turnover this implies, nor transaction costs, so the portfolio is a clean object on which to forecast risk and not an investable strategy. A buy-and-hold portfolio would have drifting weights and would give different numbers; every result in this thesis uses constant weights.

3.3 Additional samples used in the development period

The ladder of comparisons in Chapter 6 is also run on three further ETF samples: SPY, IEF and HYG; four sector funds (XLU, XLK, XLF, XLE); and four risk assets (HYG, EEM, VNQ, DBC). They were used for a check declared in the decision log before it was run (Section 4.7). The first contains a Treasury fund, whose correlation with equities falls in stress in the development data; in the other two the correlations rise in stress. HYG has a shorter history, so the two samples that contain it start forecasting in 2012. A fifth sample uses two mutual funds as proxies for stocks and long Treasuries (VFINX and VUSTX, equally weighted), with forecasts from 1992 to 2019, to see a period in which the stock–bond correlation was positive. The funds are not ETFs, they price once a day, and VUSTX tracks long Treasuries and not IEF. None of these samples is used in the hold-out, and their data stop at 31 December 2019.

3.4 Sample design

Table 3.1 sets out the periods. Each forecast for day t uses the 1,260 returns before t (Section 4.1), so every sample also needs five years of earlier returns.

The development period is where I explored. I screened ten ETF universes, ten regime definitions and two tail levels, plus the long-history sample. The headline family contains 214 tests of one comparison (D versus B, defined in Chapter 4); every specification tried is logged in `docs/decisions_log.md` and none was dropped. Results on the development period are

Table 3.1: SAMPLE PERIODS AND THEIR STATUS

Period	Forecast days	Days	Status
Development	4 Jan 2010 – 31 Dec 2019	2,516	Specifications were chosen after seeing results on these days.
of which early	4 Jan 2010 – 31 Dec 2015	1,510	Same status as the development period.
of which late (dry run)	4 Jan 2016 – 31 Dec 2019	1,006	Used to test the pipeline. Not an independent validation.
Hold-out	2 Jan 2020 – 31 Dec 2025	1,508	One run of one pre-specified test, after the specification is fixed.

Note: ETF samples that contain HYG start forecasting in 2012 and have 1,944 development days. The first estimation window for the first hold-out forecast covers 30 December 2014 to 31 December 2019. Day counts from the result files `ladder_dm_table.csv` and `routes_dm_table.csv`; the hold-out count is a count of dates in the raw file.

therefore descriptive. Anything chosen because it looked good there, including the confirmatory test itself, has no claim to be validated by those same days.

The late part of the development period (2016–2019) serves as a dry run. I ran the final code path on it to check the pipeline end to end, and its estimates enter no argument about the hypothesis. The hidden-state chain in the dry run starts cold at its first forecast day, whereas the exploration chain had been running since 2010, so the two paths differ slightly (a median absolute VaR difference of 1.3×10^{-5} to 1.4×10^{-5} in the primary arms). The hold-out run starts cold in the same way.

The hold-out is 2 January 2020 to 31 December 2025, 1,508 forecast days. It contains the 2020 crash, the 2022 stock–bond reversal that motivates the thesis and the years around it. I run the pre-specified test on it once, after the specification file is fixed (Section 5.9).

3.5 What was seen before the specification was fixed

The following was seen or used before the specification was fixed:

1. A descriptive data analysis and a data audit displayed summary statistics for 2010 to 2025. Calendar-year SPY–IEF correlations for 2010 to 2025 (Chapter 1) and reference baselines (Section 7.4) were computed after the hold-out was scored, for description only.

2. Three unit tests read the 2010–2025 pilot returns and passed; they were excluded from the

test run until the hold-out had been scored.

3. A search printed two single-day returns, for 2020-03-10 and 2023-11-01, once by accident. They were not used, and no specification depends on them.

4. No model was fitted or tuned on 2020–2025. But I know that the stock–bond correlation turned positive in 2022, and that knowledge motivated the question.

5. The choice of the confirmatory test was made from simulated power on a process in which the SPY–IEF correlation shift is largest by construction (Section 5.6).

6. All ETF data come from one vendor (above).

The specification file, the scoring code and the raw-data hash were fixed together in the annotated git tag `spec-freeze-v1` before the run, and the scorer checks that the run was made from code that the tag fixed. The tag is a self-registration in a repository, not an entry in an independent registry, and it carries no external timestamp. One later amendment of the scorer's validity rule, made before any scoring, is described in Section 7.1.

4 METHODOLOGY

This chapter specifies every forecasting model and states exactly what is estimated, when, and on which window, so that the study can be reimplemented from the text. The models share one regime signal, one portfolio and one estimation scheme, and differ in how they treat volatility, correlation and the mean. Section 4.7 then states what each comparison between two models changes. Chapter 5 defines the loss, the test and the decision rule.

4.1 The forecasting task and the estimation scheme

For each forecast day t I forecast the one-day-ahead value-at-risk VaR_t and expected shortfall ES_t of the portfolio return y_t at tail level α, using returns through day $t-1$ only. VaR_t is the α-quantile of the forecast distribution of y_t, and ES_t is the mean of y_t below that quantile. Both

are negative numbers. The primary tail level is $\alpha = 5\%$ and the secondary one is 2.5%.

Forecasts come from a walk-forward scheme. The forecast days are split into blocks of 21 consecutive days. For the block starting at day t_0, every model is estimated on the rolling window of the 1,260 returns before t_0 (about five years) and used, without re-estimation, for the 21 days of the block. The regime model is refitted at every block. The GARCH margins of Section 4.4 are refitted every third block, that is every 63 days, counting from the first block of the run; at the other blocks their parameters are kept and only the window is re-filtered. The first block of a run starts the hidden-state chain cold (Section 4.2). A model is therefore a forecasting procedure, consisting of the specification, the window length and the refit schedule, and this is the object that Chapter 5 compares.

4.2 The regime signal

The regime signal is a two-state Gaussian hidden Markov model (HMM) fitted to the SPY daily simple return only, so that the same signal feeds every model. Let $S_t \in \{1, 2\}$ be the hidden state, with state 1 the calm state (lower return variance) and state 2 the stress state. Given $S_t = k$, the SPY return is normal with mean μ_k and variance σ_k^2, and S_t is a Markov chain with transition matrix P. I estimate (μ_k, σ_k^2, P) on the window by the expectation-maximisation algorithm (`hmmlearn` 0.3.3, diagonal covariance, tolerance 10^{-4}, seed 0). The first fit of a run is a cold start (100 iterations; initial values in Appendix A) and every later fit warm-starts from the previous block's estimates with at most 50 iterations. States are ordered by variance after each fit.

Two sets of probabilities are used, and they must not be confused. Inside the window, the smoothed probabilities $\gamma_{s,k} = \Pr(S_s = k \mid \text{window})$ for each window day s weight the regime moments below; they use data up to the end of the window and nothing after it. For the forecast, the weights of the mixture are predictive probabilities

$$p_{t,k} = \Pr(S_t = k \mid \text{SPY returns through } t-1), \tag{1}$$

computed by propagating the filtered probability at the window end through P and, within the block, filtering with the realised SPY returns up to day $t-1$. Day t's own return is never used.

4.3 Static regime-covariance models: A, B and D

For weights $q_s \geq 0$ over the window days, the weighted mean and covariance of the asset returns are

$$\hat{\mu}(q) = \frac{\sum_s q_s r_s}{\sum_s q_s}, \qquad \hat{\Sigma}(q) = \frac{\sum_s q_s \, (r_s - \hat{\mu}(q))(r_s - \hat{\mu}(q))'}{\sum_s q_s}, \tag{2}$$

with the maximum-likelihood normalisation. The pooled moments use $q_s = 1$ and are written $\hat{\mu}, \hat{\Sigma}$. The regime moments use $q_s = \gamma_{s,k}$, with a floor of 10^{-12}, and are written $\hat{\mu}_k, \hat{\Sigma}_k$. Write $\hat{\Sigma} = \operatorname{diag}(\hat{\sigma})\hat{R}\operatorname{diag}(\hat{\sigma})$ and $\hat{\Sigma}_k = \operatorname{diag}(\hat{\sigma}_k)\hat{R}_k \operatorname{diag}(\hat{\sigma}_k)$ for the decompositions into standard deviations and correlations of the raw returns. The three static models assign to regime component k the covariance

$$\text{A (pooled normal):} \quad \Sigma_k^A = \hat{\Sigma}, \tag{3}$$

$$\text{B (regime volatility):} \quad \Sigma_k^B = \operatorname{diag}(\hat{\sigma}_k)\, \hat{R} \operatorname{diag}(\hat{\sigma}_k), \tag{4}$$

$$\text{D (regime volatility and correlation):} \quad \Sigma_k^D = \hat{\Sigma}_k. \tag{5}$$

Model A has the same covariance in both components, so it is a single normal distribution. Model B lets the volatilities differ by regime and holds the correlation matrix at its pooled value. Model D lets every volatility and every correlation differ by regime. The portfolio variance in component k is $s_k^2 = w'\Sigma_k w$, and every component has the same mean $m = w'\hat{\mu}$, the pooled window mean (the "pooled" mean treatment used in the ladder tables). The covariances are fixed within a 21-day block; only the mixture weights $p_{t,k}$ move from day to day. Two controls vary the mean: a regime-specific mean ($w'\hat{\mu}_k$ in component k) and a zero mean. I report them in the appendix because they change the size of some contrasts.

4.4 Models with GARCH margins: CCC, RSC and RSC_P

These models replace the block-constant volatilities by GARCH(1,1) volatilities that update daily. For each asset i let $x_{i,s} = 100\, r_{i,s}$. On the window I estimate

$$x_{i,s} = \mu_i + \varepsilon_{i,s}, \qquad \sigma^2_{i,s+1} = \omega_i + a_i\, \varepsilon^2_{i,s} + b_i\, \sigma^2_{i,s}, \tag{6}$$

by Gaussian quasi-maximum likelihood (the `arch` package, 7.2.0), one asset at a time. Given the estimates, I re-filter the window with the recursion, starting from the sample variance of $\varepsilon_{i,\cdot}$ in the window, and form the standardised residuals $z_{i,s} = \varepsilon_{i,s}/\sigma_{i,s}$. The variance for the first day of the block comes from continuing the recursion, and for later days in the block it is updated each day from the realised return with the parameters held fixed. Write $\sigma_t = (\sigma_{1,t}, \sigma_{2,t}, \sigma_{3,t})'$ for the volatility vector and $A_t = w \circ \sigma_t/100$, where $\circ$ is the element-wise product. The volatility vector is the same in every regime component: GARCH volatility in these models does not depend on the regime. The regime enters only through the correlation matrix of z and the mixture weights.

The three models differ in the correlation matrix of the standardised residuals:

$$\text{CCC:} \quad R_1 = R_2 = \bar{R}, \text{ a single component;} \tag{7}$$

$$\text{RSC:} \quad R_k = \bar{R}_k; \tag{8}$$

$$\text{RSC_P:} \quad R_k = \bar{R} \text{ with the (SPY, IEF) entry replaced by } (\bar{R}_k)_{12}. \tag{9}$$

Here $\bar{R}$ is the correlation matrix of z over the window with equal weights and $\bar{R}_k$ is the correlation matrix of z with weights $\gamma_{s,k}$. In RSC_P only the SPY–IEF correlation switches with the regime and the other two correlations stay pooled. I fixed this pair in advance from the stock–bond hypothesis, not by scanning. If a regime or hybrid matrix is not positive definite (smallest eigenvalue below 10^{-8}), it is repaired by clipping the eigenvalues and rescaling to a unit diagonal, which can change every entry. In the development data this repair never fired: 0 of 240 matrices

in the chain that starts in 2010, 0 of 96 in the chain that starts in 2016, and 0 of 2 in the window that ends on 31 December 2019. Every repair and every failed margin fit in the hold-out run is counted and reported (Chapter 7).

The forecast distribution of y_t for these models is a mixture of normals with weights $p_{t,k}$ ($p_{t,k} = 1$ for CCC), component means all equal to $m = w'\hat{\mu}^{\mathrm{QML}}/100$ (the QML constants), and component variances $A_t' R_k A_t$. The mean is the same in the two models of the primary test and is not regime-specific.

4.5 Mixture value-at-risk and expected shortfall

For components with weights p_k, means m_k and standard deviations s_k, the value-at-risk is the solution q of

$$\sum_k p_k \, \Phi\!\left(\frac{q - m_k}{s_k}\right) = \alpha, \tag{10}$$

found by root-finding, and the expected shortfall is

$$\mathrm{ES} = \frac{1}{\alpha} \sum_k p_k \Big[m_k \, \Phi(z_k) - s_k \, \phi(z_k) \Big], \qquad z_k = \frac{q - m_k}{s_k}, \tag{11}$$

where Φ and ϕ are the standard normal distribution and density functions. Both are returns, so both are negative at the levels used here. This is the Gaussian chain, which is the primary one. In the secondary Student-t chain every component is a Student-t distribution with common degrees of freedom ν, scaled to the component's standard deviation and estimated by maximum likelihood on the window's portfolio returns standardised by the model's own in-window mixture mean and standard deviation; ν is therefore not comparable across the static and GARCH families.

4.6 Benchmarks

Two other models serve as reference points. The first is a dynamic conditional correlation model (DCC; Engle, 2002) built on the same GARCH(1,1) margins: the correlation matrix follows

$Q_t = (1 - \theta_1 - \theta_2)\bar{Q} + \theta_1 z_{t-1} z'_{t-1} + \theta_2 Q_{t-1}$, normalised to a correlation matrix, with (θ_1, θ_2) estimated on the window by Gaussian quasi-maximum likelihood every 63 days and the recursion updated daily in between. If the optimiser returns $\theta_1 < 0$, $\theta_2 < 0$ or $\theta_1 + \theta_2 \geq 0.9995$, the code sets $\theta_1 = \theta_2 = 0$, and the number of such fallbacks is counted. The second is a GARCH(1,1) model with Student-t errors fitted by maximum likelihood to the portfolio return itself (GARCH-t), which uses no information about the individual assets. Two further reference rows are used only post hoc (Section 7.4), both on the portfolio return: historical simulation, the empirical 5% quantile and the mean of the returns below it over the last 500 days; and a RiskMetrics exponentially weighted variance with decay 0.94, a zero mean and a normal distribution.

4.7 What each comparison changes

The thesis compares models in pairs that share the data, the regime signal and the estimation window. I do not call these pairs a nested sequence, because several pairs differ in more than one respect. Table 4.1 lists, for each comparison, the richer model and the baseline, what differs between them, and whether one model is a special case of the other. I derived each cell from the code (`src/explore_wf.py`, `src/explore_routes.py`, `src/explore_ladder.py`).

Three consequences follow. First, only the contrasts D versus B, RSC versus CCC and RSC_P versus CCC change one thing, the correlation matrix, with the volatility model fixed. They are the only ones that support a statement about the correlation channel, and the statement holds only within the volatility model used. Second, the contrasts CCC versus B and RSC versus D change the volatility model, the correlation input, the mean estimator and the number of components together. They compare two classes of model, and they cannot be described as the effect of volatility dynamics: the GARCH models used here also remove the regime dependence from volatility. Third, because the margins of CCC, RSC and RSC_P are the same, the primary test of the thesis, RSC_P versus CCC, differs in two respects only: a mixture of two components against one, and a regime-specific SPY–IEF correlation entry against a pooled one.

Two families of comparisons were declared before they were scored. The first is the volatility-

Table 4.1: WHAT DIFFERS BETWEEN THE TWO MODELS IN EACH COMPARISON

Comparison (richer vs baseline)	Volatility	Correlation matrix	Mean	Components	Relation
B vs A	regime-specific block SDs vs one pooled SD	pooled raw-return correlation in both	pooled window mean	2 vs 1	Nested in volatility
D vs B	same regime SDs	regime-specific vs pooled, raw returns	pooled window mean	same	Nested in the correlation matrix
CCC vs B	single-regime GARCH(1,1), daily update, vs regime-specific block SDs	standardised residuals vs raw returns	QML constant vs window mean	1 vs 2	Non-nested: changes four things
RSC vs D	single-regime GARCH(1,1) vs regime-specific block SDs	regime-specific, standardised residuals vs raw returns	QML constant vs window mean	2 and 2	Non-nested: model-class comparison
RSC vs CCC	identical GARCH	regime-specific vs pooled, same z	identical	2 vs 1	Nested in the correlation matrix
RSC_P vs CCC (primary)	identical GARCH	only the (SPY, IEF) entry regime-specific	identical	2 vs 1	Nested in one correlation entry
DCC vs CCC	same specification, fitted separately; own variance initialisation	recursion updated daily vs one pooled matrix per block	identical	1 vs 1	Parameter-nested ($\theta_1 = \theta_2 = 0$), different code path
DCC vs RSC	as above	smooth recursion vs regime-weighted	identical	1 vs 2	Non-nested

Note: "Block SDs" means standard deviations estimated on the window and held fixed for 21 days. B reduces to A when the two regime SDs are equal; D reduces to B and RSC to CCC when the regime correlation matrices equal the pooled one. Static arms use the pooled window sample mean and GARCH arms the QML constant mean. The DCC filter starts from the `arch` package's own variance initialisation and carries its correlation state forward daily, so it is not the CCC fit with a constant added.

model contrast RSC versus D with a pooled mean, in the four ETF samples, with Holm's adjustment over the four (a gate: the ladder would be the main thesis if the contrast were negative in all four and significant after adjustment in at least two; Section 6.3 reports that it was not). The second is the confirmatory test RSC_P versus CCC (Chapter 5).

5 EVALUATION AND INFERENCE

This chapter defines the loss, the estimand, the test, the decision rule and the power study, and states which results are pre-specified and which are not.

5.1 The loss function

A forecast of $(\mathrm{VaR}_t, \mathrm{ES}_t)$ is scored against the realised return y_t with the loss

$$\mathrm{FZ0}(y_t, \mathrm{VaR}_t, \mathrm{ES}_t; \alpha) = -\frac{\mathbf{1}\{y_t \le \mathrm{VaR}_t\}\,(\mathrm{VaR}_t - y_t)}{\alpha\,\mathrm{ES}_t} + \frac{\mathrm{VaR}_t}{\mathrm{ES}_t} + \log(-\mathrm{ES}_t) - 1, \tag{12}$$

where $\mathrm{VaR}_t < 0$ and $\mathrm{ES}_t < 0$ are return quantiles. Lower is better. This is the member of the Fissler–Ziegel family (Fissler and Ziegel, 2016) that Patton et al. (2019) call FZ0; Joint elicitability means that the expected loss is smallest at the true VaR and ES pair, which is the reason to use such a loss.[1] Scaling the return and both forecasts by $c > 0$ adds $\log c$ to the loss, so a difference between the losses of two models does not depend on the units of the return. Section 5.7 translates a loss difference into a proportional error in the forecasts.

5.2 Estimand

Let model R (richer) and model B (baseline) be two forecasting procedures, each a model together with its window and refit schedule (Section 4.1). Define the loss difference on forecast day t as

$$d_t = \mathrm{FZ0}_t^R - \mathrm{FZ0}_t^B, \tag{13}$$

so that a negative d_t favours the richer model. This sign convention, richer minus baseline with negative meaning better, is used in every table. The estimand is the long-run mean

$$\mu = \mathbb{E}[d_t], \tag{14}$$

the expected loss difference between the two procedures over the evaluation sample. It describes the procedures as they are run, with estimation error in their parameters included. It is not a statement about whether the extra parameters of the richer model are zero in a population. This

[1] I did not re-derive the elicitability result and rely on the cited papers for it. I use the loss as a formula; a unit test checks that on simulated normal data the true VaR and ES have lower mean loss than biased ones.

is the sense of equal predictive ability in Giacomini and White (2006), who treat the forecasting method, including its finite estimation window, as the object being compared, so that estimation uncertainty does not vanish and the test applies whether or not one model is a special case of the other. Their unconditional test is the Diebold–Mariano statistic with a two-sided alternative; the one-sided version used here is my adaptation. McCracken (2020) gives a counterexample to asymptotic normality for a fixed estimation window, and the rolling-window case of Giacomini and White, which my design uses (1,260 returns), remains valid.

5.3 Estimator and test

The estimator of μ is the sample mean $\bar{d} = T^{-1}\sum_{t=1}^{T} d_t$ over the T forecast days of the evaluation sample. Its variance is estimated with a Bartlett-kernel heteroskedasticity- and autocorrelation-consistent (HAC) estimator,

$$\widehat{V} = \hat{\gamma}_0 + 2\sum_{\ell=1}^{m}\Big(1 - \frac{\ell}{m+1}\Big)\hat{\gamma}_\ell, \qquad \hat{\gamma}_\ell = \frac{1}{T}\sum_{t=\ell+1}^{T}(d_t - \bar{d})(d_{t-\ell} - \bar{d}), \qquad \widehat{\mathrm{SE}} = \sqrt{\widehat{V}/T}, \tag{15}$$

with the lag $m = \lfloor 4\,(T/100)^{2/9} \rfloor$, the rule of thumb associated with Newey and West (1994) and not their data-driven selection. At $T = 1{,}508$ this gives $m = 7$. The test statistic is

$$t_{\mathrm{DM}} = \bar{d}/\widehat{\mathrm{SE}}, \tag{16}$$

which I compare with the standard normal distribution, without a small-sample correction (the small-sample correction of Harvey et al. (1997) has factor 0.9997 at one step and $T = 1{,}508$, and I do not apply it). This is the Diebold–Mariano statistic (Diebold and Mariano, 1995). The null hypothesis and its alternative are statements about μ and are written separately from the statistic:

$$H_0 : \ \mu \geq 0 \qquad \text{against} \qquad H_1 : \ \mu < 0. \tag{17}$$

In words, under H_0 the richer model does not have a lower expected loss than the baseline. The one-sided p-value is $\Phi(t_{\mathrm{DM}})$ and the two-sided p-value is $2\{1-\Phi(|t_{\mathrm{DM}}|)\}$. Every table gives the two-sided 95% interval $\bar{d} \pm 1.96\,\widehat{\mathrm{SE}}$ and labels each p-value one- or two-sided.

Normal critical values need not be accurate for out-of-sample comparisons. When one model is a special case of the other and the null is that the smaller model is correct, the limiting distributions are non-standard (Clark and McCracken, 2001), and Clark and West (2007) give an adjusted statistic whose normal critical values give sizes close to nominal. Both papers treat linear models scored by squared error, so the results do not carry over to the mixture forecasts and loss used here. For a rolling window of fixed length, Clark and West (2007) themselves read the usual t-statistic as a statement about coverage for the estimated-parameter mean difference, which is the reading used in this thesis. I therefore read the size of the test from simulation (Section 5.6) and not from asymptotic theory.

5.4 Decision rule and interpretation

The confirmatory test is RSC_P against CCC, at tail level $\alpha = 5\%$ with a Gaussian chain (Section 4.7). Two levels should not be confused: $\alpha = 5\%$ is the tail level of the forecast, and the test is a separate test with its own size, also 5%. The decision rule is one-sided: reject H_0 if $t_{\mathrm{DM}} < -1.6449$. I chose a one-sided test because the substantive question has one direction, and because the simulated power and size reported in Chapter 6 were computed for exactly this rule. A one-sided test says nothing about harm, so I declare a separate statement: the switching model is called significantly worse if $t_{\mathrm{DM}} > +1.96$, the upper tail of a two-sided 5% test. In the simulated null the upper-tail rate of this comparison is 5.2% at 1.96 (Table 6.6) and 8.0% at 1.6449, which is why 1.96 is the threshold.

The four possible outcomes are checked in this order and each licenses one sentence, fixed before the run. The full wording of O1, O3 and O4 is in Appendix A.

O1 $t_{\mathrm{DM}} \leq -1.6449$: statistical evidence at the one-sided 5% level that switching lowers expected FZ0 loss.

O4 $t_{\mathrm{DM}} \geq +1.96$: evidence that the switching specification has higher expected FZ0 loss.

O3 otherwise, and $\bar{d} - 1.6449\,\widehat{\mathrm{SE}} > -0.002$: the data are inconsistent, at the one-sided 5% level, with an improvement larger than SESOI $= 0.002$ FZ0 units.

O2 otherwise: "The hold-out data neither establish an improvement nor exclude one. The estimate is $\bar{d}$ (95% CI $[a, b]$); an improvement of at least M (the minimum detectable difference for the one-sided 5% test, 80% power) would have been detected with 80% probability, and smaller improvements, including none, are consistent with the data."

No outcome is worded as an absence of an effect. The interval and the minimum detectable difference are reported whatever the outcome.

5.5 Minimum detectable difference and power

The minimum detectable difference (MDE) is the smallest true difference that a test detects with 80% probability, given its standard error. For the two-sided 5% test it is $(1.96 + 0.8416)\,\widehat{\mathrm{SE}} = 2.8016\,\widehat{\mathrm{SE}}$, and for the one-sided 5% test it is $(1.6449{+}0.8416)\,\widehat{\mathrm{SE}} = 2.4865\,\widehat{\mathrm{SE}}$. Every comparison table gives the two-sided value, labelled as such, and the primary test gives both. The MDE depends on the design (the sample length and the variability of the loss difference) and not on the estimate, so it can be reported next to a result. I never report power computed at the observed estimate: that quantity is a function of the p-value and adds no information (Hoenig and Heisey, 2001). Power is reported only for stated effect sizes, which are the simulated scenarios below, the smallest effect of interest and the MDE.

5.6 The power study

I simulate a stylised process with the structure of the problem and score the same pipeline on it. A two-state Markov chain (calm persistence 0.99, stress persistence 0.97) drives three assets, ordered as an equity-like, a bond-like and a gold-like asset. Calm volatilities are 0.75%, 0.30% and 0.90% per day and stress volatilities are 2.0%, 0.5% and 1.6%. Innovations are multivariate Student-t with six degrees of freedom. Each asset's volatility also follows GARCH(1,1) dynamics

(coefficients 0.08 and 0.90 on the regime-scaled return, fixed and not estimated). The calm correlations of (equity–bond, equity–gold, bond–gold) are $(-0.30, 0.05, 0.30)$. The four scenarios set the stress correlations: S0 leaves them unchanged (the null for the correlation steps); S1, a moderate breakdown, sets $(0.00, 0.20, 0.30)$; S2, a strong breakdown, sets $(0.40, 0.40, 0.30)$; S3, a flight to quality, sets $(-0.55, -0.10, 0.40)$. Each replication simulates 1,260 burn-in days and 1,508 test days (the length of the hold-out), runs the whole pipeline of Chapter 4 including the HMM on the first asset, and applies the test of Section 5.3. There are 500 replications per scenario, with seeds distinct from an earlier study that used 2,520 test days. Size is the rejection rate in S0. Power is the rejection rate in S1 to S3, with a binomial 95% interval.

This is an assumption and not a measurement. The process, its parameters and the GARCH coefficients are not estimated from data, and the shift in stress correlation is largest in the stock–bond pair by construction. The choice of RSC_P as the confirmatory test rested on an earlier run of this simulation, in which switching only the SPY–IEF correlation had equal size and higher power than switching all correlations. That advantage reflects the design of the scenarios and is not a finding about markets. The decision was made before the hold-out was opened and it is not reopened here, whatever the 1,508-day study shows.

5.7 The smallest effect of interest

A non-significant result does not show that an effect is absent, and the minimum detectable difference says only what the design could find. To give a negative result some content I declare one reference value, the smallest effect of interest (SESOI), and use it as a one-sided bound. I translate it into economic terms by asking what loss difference a given forecast error produces. Suppose a forecaster's VaR and ES both equal a factor c times the true values of a known distribution. The expected loss exceeds that of the true forecast by a penalty that is approximately proportional to $(c-1)^2$. Table 5.1 gives the penalty computed by numerical integration for a unit-variance normal and a Student-t distribution with six degrees of freedom (`scripts/confirm_04_sesoi.py`; no market data are used). For the normal at $\alpha = 5\%$,

Table 5.1: EXPECTED FZ0 PENALTY OF SCALING BOTH VAR AND ES BY A CONSTANT FACTOR

True distribution	$c = 0.95$	$c = 0.97$	$c = 1.03$	$c = 1.05$
Normal, $\alpha = 5\%$	0.0051	0.0018	0.0016	0.0042
Student-$t(6)$, $\alpha = 5\%$	0.0040	0.0014	0.0013	0.0034
Normal, $\alpha = 2.5\%$	0.0067	0.0023	0.0020	0.0055

Note: Analytic expected loss at scale factor c minus the expected loss at $c = 1$, for a unit-variance true distribution; no market data are used. Source: `output/confirm/sesoi.json`.

understating or overstating both forecasts by 3% costs about 0.0016 to 0.0018. I therefore declare SESOI = 0.002, one significant figure of the average of these two penalties, as the difference corresponding to a proportional forecast-scale error of roughly 3%. That a 3% error in a capital-relevant quantity is economically meaningful is my assumption and not a fact. With the Student-t distribution the same penalty is about 20% lower and rounds to 0.001; I report this as a caveat and keep the declared value.

The reference value is used in outcome O3 only. With a standard error of 0.0010 to 0.0015, the range suggested by the development data, the one-sided bound $\bar{d} - 1.6449\,\widehat{\mathrm{SE}}$ lies 0.0021 to 0.0025 below the estimate, so O3 requires an estimate near zero or positive and is unlikely. I state this in advance. A two-one-sided-tests equivalence procedure with margin 0.002 is out of reach, since the half-width of its 90% interval alone, 0.0021 to 0.0025, would exceed the margin, so the thesis does not claim equivalence under any outcome.

5.8 Calibration tests

Next to the loss, I report whether the forecasts have the right coverage, as description and not as part of the decision. The hit rate is the share of days on which $y_t \leq \mathrm{VaR}_t$; it should be near α. The Kupiec (1995) test compares the hit rate with α through a likelihood ratio with a χ^2_1 reference. The Christoffersen (1998) independence test asks whether hits cluster in time, using a first-order Markov alternative. Their sum, $LR_{uc} + LR_{ind}$ with a χ^2_2 reference, is the

conditional-coverage test. For expected shortfall I use the statistic

$$Z_2 = \frac{1}{T\alpha} \sum_{t=1}^{T} \frac{\mathbf{1}\{y_t \le \mathrm{VaR}_t\}\, y_t}{\mathrm{ES}_t} - 1, \tag{18}$$

which is zero in expectation if the forecast distribution is correct and positive if risk is underestimated. It is the statistic that Acerbi and Székely (2014) call Z_2 with its sign reversed, because they write the expected shortfall as a positive loss. Its expectation result does not require independent observations. I obtain its p-value by their simulation approach: I draw 5,000 simulated samples, each with one return per forecast day drawn from that day's forecast mixture, compute Z_2 for each, and report the share at or above the observed value. A unit test on synthetic data checks the sign convention.

5.9 Multiple testing and what is pre-specified

The confirmatory family has one member, so no adjustment is needed. Besides it I score six secondary analyses of the same forecasts, none of which adds a model, a sample or a regime signal: the six ladder steps at the hold-out; the calendar year 2022 (interval only); a Giacomini–White conditional test of the primary pair with the instruments {1, lagged 63-day SPY–IEF correlation} (a Wald statistic with a χ^2_2 reference, read with caution because Zhu and Timmermann (2020), in a preprint, show that the conditional null can fail when forecasts rest on estimated parameters); the same comparison at $\alpha = 2.5\%$ and in the Student-t chain; a test of superior predictive ability against GARCH-t (Hansen, 2005; stationary bootstrap, 2,000 draws, seed 0; candidates B, D, RSC, RSC_P); and the calibration tests above. They are unadjusted and labelled secondary, and none is cited as confirmation. Anything else computed is marked POST HOC and carries no inferential weight.

In the development data, the 214 headline tests of the exploration were treated as one family with the adjustment of Holm (1979), and the pre-declared family of four ETF samples with another. The static test of D versus B is oversized under GARCH margins (Chapter 6), so

its adjusted values are not valid; nothing was significant after adjustment, so the reading is unchanged.

The specification is registered in `docs/preregistration_confirmatory.md` and fixed by an annotated git tag together with the scoring code and the raw-data hash. This follows the logic of pre-registration (Nosek et al., 2018) but it is a self-registration inside the repository, with no independent timestamp. The run followed the protocol except for one amendment of the scorer's validity rule, made before any scoring (Section 7.1).

5.10 Limits of the design

The following limits apply to every result in this thesis and are not repeated in each table.

- The test compares forecasting procedures, with their window, refit schedule and the hidden-state chain. It does not test population parameters.
- The uncertainty in the estimated regimes is not carried into the test. I did not build a bootstrap of the whole pipeline; the rejection rate in the simulated null stands in as a size check, and it is only as good as the simulated process.
- The simulated power does not transfer one-for-one to real data: the simulated standard error of the primary difference depends on the scenario and differs from the real-data one (Chapter 6), so the power in a scenario is not the power that the same difference would have at the standard error of the real data.
- The regime signal is the SPY return only. A volatility signal and a rolling 1,260-day window may be slow to reflect a change in the stock–bond correlation.
- There is one portfolio of three funds, one data vendor and one reversal episode in the hold-out.

6 RESULTS: DEVELOPMENT DATA AND POWER

This chapter reports what the development data (forecast days through 2019) show about where regime information helps, and how large an effect the design could reveal. It does not report the hold-out. The development results are descriptive: I examined ten ETF universes, ten regime definitions and two tail levels, and chose specifications after seeing them (Chapter 3). The comparisons below were run through the ladder of Chapter 4, and each states what it changes (Table 4.1).

6.1 How to read the tables

Every comparison reports the mean FZ0 difference $\bar{d}$ of the richer model minus the baseline, so negative favours the richer model; the HAC standard error; the two-sided 95% interval; the two-sided p-value; and the minimum detectable difference (MDE), the smallest true difference a two-sided 5% test detects with 80% probability. A result inside the MDE is inconclusive for effects below the MDE; it does not show that the effect is absent. The tail level is 5%, the chain Gaussian and the mean pooled unless stated.

6.2 Regime-specific volatility against a pooled normal

Table 6.1 compares model B, which gives each regime its own volatilities, with the pooled normal A. In three of the four ETF samples the improvement is large, each with a two-sided p-value below 0.001, and it is also large in the fund sample. The exception is the main sample, SPY/IEF/GLD, where the estimate is -0.0201 (two-sided $p = 0.408$) with an MDE of 0.0679, so its interval, from -0.0675 to 0.0274, includes both a large improvement and none. Giving the volatility a regime dependence therefore lowers the loss substantially in three of four ETF samples, and this development sample cannot show it in the fourth.

Table 6.1: REGIME-SPECIFIC VOLATILITY AGAINST A POOLED NORMAL (B VERSUS A)

Sample	n	$\bar{d}$	SE	95% interval	t	p (2-sided)	MDE
B versus A							
SPY/IEF/GLD	2,516	−0.0201	0.0242	[−0.0675, 0.0274]	−0.83	0.408	0.0679
SPY/IEF/HYG	1,944	−0.1659	0.0373	[−0.2390, −0.0929]	−4.45	<0.001	0.1044
Sectors	2,516	−0.1806	0.0252	[−0.2300, −0.1312]	−7.17	<0.001	0.0705
Risk assets	1,944	−0.1586	0.0217	[−0.2010, −0.1161]	−7.32	<0.001	0.0607
Funds 1992–2019	7,053	−0.1057	0.0293	[−0.1630, −0.0483]	−3.61	<0.001	0.0820

Note: Mean FZ0 difference $\bar{d}$ of the richer model minus the baseline over the forecast days, tail level $\alpha = 5\%$, Gaussian chain, pooled mean; negative favours the richer model. SE: Bartlett-kernel HAC standard error. The interval is $\bar{d} \pm 1.96$ SE. The p-value is two-sided. MDE $= 2.8016$ SE is the smallest true difference a two-sided 5% test detects with 80% power. Source: `output/explore/ladder/ladder_dm_table.csv`. Forecast days 2010–2019 (ETFs; credit and risk-asset samples start in 2012), 1992–2019 (funds).

6.3 A single-regime GARCH model class against static regime covariance

Table 6.2 and Figure 6.1 turn to the contrasts that replace block-constant regime volatilities by GARCH(1,1) volatilities. These contrasts compare model classes: the GARCH models have no regime dependence in volatility and also change the correlation input and the mean estimator (Table 4.1). The pre-declared family is RSC versus D in the four ETF samples. The estimates are negative in all four. After Holm's adjustment over the four, the adjusted p-values are 0.2341, 0.2341, 0.2247 and 0.0025. The pre-declared gate required significance after adjustment in at least two samples; it was met in one (the risk assets). The result is therefore reported as suggestive and not as established. I did not reframe the thesis around it.

The size of the contrast depends on the mean treatment, though not its sign. With a zero mean in both models the RSC-versus-D estimates range from −0.0339 to −0.0814, larger in absolute value than with the pooled mean; with regime-specific means they are close to the pooled ones (−0.0263 in the main sample). I therefore state the contrast with the mean treatment attached, and the zero-mean and regime-mean tables are in the appendix.

Calibration points the same way (Table 6.3). The static models overstate risk on the ETF samples: the 5% VaR is breached on only 2.8% to 4.2% of days for A, B and D, while the GARCH-margin models breach it on 4.3% to 5.5%. In the fund sample the static models are

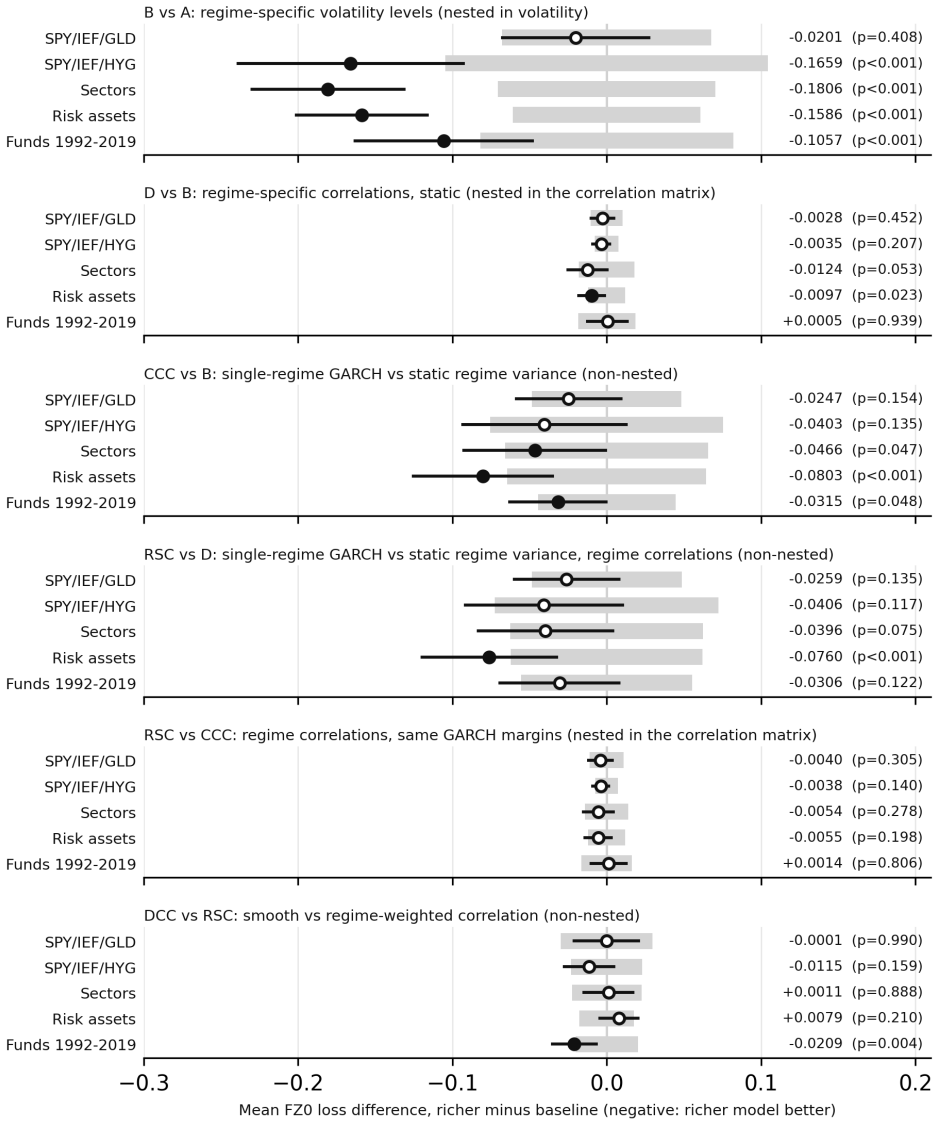


Figure 6.1: MEAN FZ0 DIFFERENCE, RICHER MINUS BASELINE, FOR EACH COMPARISON AND SAMPLE

Note: Points are $\bar{d}$; lines are two-sided 95% intervals; the grey band is $\pm$ the MDE; filled points have a two-sided $p < 0.05$ (unadjusted). Negative favours the richer model. Tail level 5%, Gaussian chain, pooled mean. Source: `output/explore/ladder/ladder_dm_table.csv`, drawn by `scripts/thesis_fig_forest.py`.

Table 6.2: GARCH MARGINS AGAINST STATIC REGIME COVARIANCE

Sample	n	$\bar{d}$	SE	95% interval	t	p (2-sided)	MDE	Holm p
CCC versus B (non-nested)								
SPY/IEF/GLD	2,516	−0.0247	0.0173	[−0.0586, 0.0093]	−1.43	0.154	0.0485	
SPY/IEF/HYG	1,944	−0.0403	0.0269	[−0.0931, 0.0125]	−1.49	0.135	0.0755	
Sectors	2,516	−0.0466	0.0235	[−0.0926, −0.0006]	−1.99	0.047	0.0658	
Risk assets	1,944	−0.0803	0.0230	[−0.1254, −0.0351]	−3.49	<0.001	0.0645	
Funds 1992–2019	7,053	−0.0315	0.0159	[−0.0627, −0.0003]	−1.98	0.048	0.0446	
RSC versus D (non-nested; pre-declared family)								
SPY/IEF/GLD	2,516	−0.0259	0.0173	[−0.0599, 0.0081]	−1.49	0.135	0.0486	0.2341
SPY/IEF/HYG	1,944	−0.0406	0.0259	[−0.0914, 0.0102]	−1.57	0.117	0.0726	0.2341
Sectors	2,516	−0.0396	0.0223	[−0.0832, 0.0040]	−1.78	0.075	0.0623	0.2247
Risk assets	1,944	−0.0760	0.0222	[−0.1195, −0.0325]	−3.43	<0.001	0.0622	0.0025
Funds 1992–2019	7,053	−0.0306	0.0198	[−0.0693, 0.0081]	−1.55	0.122	0.0554	–

Note: Mean FZ0 difference $\bar{d}$ of the richer model minus the baseline over the forecast days, tail level $\alpha = 5\%$, Gaussian chain, pooled mean; negative favours the richer model. SE: Bartlett-kernel HAC standard error. The interval is $\bar{d} \pm 1.96\,\mathrm{SE}$. The p-value is two-sided. MDE $= 2.8016\,\mathrm{SE}$ is the smallest true difference a two-sided 5% test detects with 80% power. Source: `output/explore/ladder/ladder_dm_table.csv`. Forecast days 2010–2019 (ETFs; credit and risk-asset samples start in 2012), 1992–2019 (funds). Holm p: the pre-declared family is RSC versus D in the four ETF samples, Holm-adjusted over the four; the fund sample is not in the family.

Table 6.3: SHARE OF DAYS ON WHICH THE RETURN FELL BELOW THE FORECAST VAR (NOMINAL 5%), DEVELOPMENT DATA

Sample	A	B	D	CCC	RSC	DCC
SPY/IEF/GLD	3.1%	3.7%	3.5%	5.0%	4.7%	5.0%
SPY/IEF/HYG	3.4%	4.2%	4.0%	5.5%	5.5%	5.0%
Sectors	3.1%	3.4%	3.7%	5.2%	5.4%	5.5%
Risk assets	2.8%	2.8%	3.0%	4.3%	4.6%	4.7%
Funds 1992–2019	4.8%	5.4%	5.3%	5.5%	5.2%	5.5%

Note: Gaussian chain, pooled mean, forecast days as in Table 6.1. A hit is a day with $y_t \le \mathrm{VaR}_t$. Source: `output/explore/ladder/ladder_loss_table.csv`.

close to nominal (B 5.4%, D 5.3%), so this statement is about the ETF samples.

6.4 The correlation steps

Table 6.4 reports the steps that change only the correlation matrix. With GARCH margins fixed, letting all correlations switch (RSC versus CCC) gives small negative estimates in the four ETF samples (−0.0040 to −0.0055) and +0.0014 in the funds. Their MDEs are 0.0073 to 0.0162, and every interval includes zero (two-sided p-values from 0.140 to 0.806). I do not detect a gain from switching correlations in these samples. The data are consistent with gains up to about the MDE, and the estimates are between about a third and a half of it (0.36 to 0.52). The same comparison with a zero mean gives −0.0087 in the sector sample (two-sided $p = 0.046$,

Table 6.4: CORRELATION STEPS

Sample	n	$\bar{d}$	SE	95% interval	t	p (2-sided)	MDE
D versus B (nested in the correlation matrix; size unreliable under GARCH margins)							
SPY/IEF/GLD	2,516	−0.0028	0.0037	[−0.0101, 0.0045]	−0.75	0.452	0.0104
SPY/IEF/HYG	1,944	−0.0035	0.0028	[−0.0089, 0.0019]	−1.26	0.207	0.0078
Sectors	2,516	−0.0124	0.0064	[−0.0249, 0.0002]	−1.93	0.053	0.0179
Risk assets	1,944	−0.0097	0.0043	[−0.0182, −0.0013]	−2.27	0.023	0.0120
Funds 1992–2019	7,053	0.0005	0.0066	[−0.0125, 0.0135]	0.08	0.939	0.0185
RSC versus CCC (nested in the correlation matrix)							
SPY/IEF/GLD	2,516	−0.0040	0.0039	[−0.0117, 0.0037]	−1.02	0.305	0.0110
SPY/IEF/HYG	1,944	−0.0038	0.0026	[−0.0089, 0.0013]	−1.48	0.140	0.0073
Sectors	2,516	−0.0054	0.0050	[−0.0151, 0.0044]	−1.08	0.278	0.0139
Risk assets	1,944	−0.0055	0.0043	[−0.0139, 0.0029]	−1.29	0.198	0.0120
Funds 1992–2019	7,053	0.0014	0.0058	[−0.0099, 0.0128]	0.25	0.806	0.0162
DCC versus RSC (non-nested)							
SPY/IEF/GLD	2,516	−0.0001	0.0106	[−0.0209, 0.0207]	−0.01	0.990	0.0297
SPY/IEF/HYG	1,944	−0.0115	0.0082	[−0.0276, 0.0045]	−1.41	0.159	0.0229
Sectors	2,516	0.0011	0.0081	[−0.0147, 0.0169]	0.14	0.888	0.0226
Risk assets	1,944	0.0079	0.0063	[−0.0045, 0.0203]	1.25	0.210	0.0177
Funds 1992–2019	7,053	−0.0209	0.0072	[−0.0351, −0.0068]	−2.90	0.004	0.0202

Note: Mean FZ0 difference $\bar{d}$ of the richer model minus the baseline over the forecast days, tail level $\alpha = 5\%$, Gaussian chain, pooled mean; negative favours the richer model. SE: Bartlett-kernel HAC standard error. The interval is $\bar{d} \pm 1.96$ SE. The p-value is two-sided. MDE $= 2.8016$ SE is the smallest true difference a two-sided 5% test detects with 80% power. Source: `output/explore/ladder/ladder_dm_table.csv`. Forecast days 2010–2019 (ETFs; credit and risk-asset samples start in 2012), 1992–2019 (funds).

unadjusted) and −0.0070 in the risk-asset sample ($p = 0.080$), so the answer for these samples is not the same under every mean treatment. A smooth alternative (DCC versus RSC) has two-sided p-values from 0.159 to 0.990 in the ETF samples. In the fund sample DCC is better than RSC by 0.0209 ($p = 0.004$, unadjusted).

The static step D versus B needs a caution. Under the simulated process with GARCH margins and no change in correlation (scenario S0), the one-sided test of D against B rejects in 9.6% of replications (95% interval 7.0% to 12.2%) at 1,508 test days, in 11.0% (6.7% to 15.3%) at 2,520 days, and in 14.8% and 17.0% of replications at $|t| > 1.96$ (Table 6.6). The test is oversized when volatility clusters. In the exploration grid, 37 of the 214 headline D-versus-B tests have a raw $p < 0.05$ (17.3%), 18 favouring D and 19 favouring B, and none survives a Holm adjustment (smallest adjusted $p = 0.1413$). That rate matches the simulated null rate, so the raw count is not evidence of an effect, and the adjusted values are not valid for these

Table 6.5: SINGLE-PAIR CORRELATION SWITCHING AGAINST CONSTANT CORRELATION, DEVELOPMENT DATA (RSC_P VERSUS CCC)

α	Forecast days	n	$\bar{d}$	SE	95% interval	t	p (2-sided)	MDE
5.0%	2010–2019	2,516	−0.0015	0.0010	[−0.0034, 0.0005]	−1.49	0.136	0.0028
5.0%	2010–2015	1,510	−0.0012	0.0013	[−0.0038, 0.0014]	−0.89	0.371	0.0037
5.0%	2016–2019	1,006	−0.0019	0.0015	[−0.0048, 0.0010]	−1.29	0.198	0.0041
2.5%	2010–2019	2,516	−0.0028	0.0015	[−0.0058, 0.0002]	−1.82	0.069	0.0043
2.5%	2010–2015	1,510	−0.0029	0.0021	[−0.0070, 0.0012]	−1.37	0.171	0.0059
2.5%	2016–2019	1,006	−0.0027	0.0022	[−0.0070, 0.0017]	−1.20	0.231	0.0062

Note: RSC_P_N minus CCC_S_N, Gaussian chain, pooled mean; negative favours switching the SPY–IEF correlation. SE is MDE/2.8 (the table stores MDE rounded to 2.8 SE); the interval is $\bar{d} \pm 1.96$ SE. These are development data: the specification was chosen after seeing them, so they are not validation. Source: `output/explore/routes/routes_dm_table.csv`, provider HMM_SPY2.

p-values. The simulated process is stylised and the 214 tests are not independent, so the match is suggestive, not proof.

6.5 The primary pair on the development data

Table 6.5 gives the confirmatory comparison, RSC_P against CCC, on the development data. Over 2010 to 2019 the estimate is −0.0015 (two-sided $p = 0.136$, MDE 0.0028); in the early part, 2010 to 2015, it is −0.0012 ($p = 0.371$), and in the dry-run part, 2016 to 2019, −0.0019 ($p = 0.198$). The implied standard errors, 0.0010, 0.0013 and 0.0015, are the ones relevant to the hold-out, which has 1,508 days. These estimates are small and consistent with a zero difference and with a gain of the size of the SESOI of 0.002; they do not support a claim either way, and because the pair, the model and the test were chosen with these days in view they are not validation.

6.6 Power at the hold-out length

Table 6.6 shows the simulated size and power of the primary test at 1,508 test days. The size, the rejection rate in S0, is 3.6% (95% interval 2.0% to 5.2%), so there is no sign of oversizing, although 500 replications cannot show that the size is at most 5%, and the mean difference in S0 is slightly positive, so the null is interior and not on the boundary. The power for the primary test is 18.2% (14.8% to 21.6%) in the moderate scenario S1 and 35.2% (31.0% to 39.4%) in the strong scenario S2, and 2.4% in the flight-to-quality scenario S3, where correlations move

Table 6.6: SIMULATED SIZE AND POWER OF THE DIFFERENCE TESTS (500 REPLICATIONS PER SCENARIO AT 1,508 DAYS; 200 AT 2,520 DAYS)

Scenario	One-sided rejection [95% CI]	Two-sided	Upper tail	Mean $\bar{d}$	Mean SE
Primary test: RSC_P versus CCC, 1,508 test days					
S0 (no correlation change)	3.6% [2.0%, 5.2%]	7.4%	5.2%	0.0002	0.0006
S1 (moderate)	18.2% [14.8%, 21.6%]	14.2%	0.4%	−0.0009	0.0010
S2 (strong)	35.2% [31.0%, 39.4%]	24.2%	0.0%	−0.0034	0.0026
S3 (flight to quality)	2.4% [1.1%, 3.7%]	4.8%	3.4%	0.0002	0.0017
Same test, 2,520 test days (earlier study)					
S0 (no correlation change)	4.5% [1.6%, 7.4%]	7.0%	4.5%	0.0001	0.0005
S1 (moderate)	27.0% [20.8%, 33.2%]	17.5%	0.0%	−0.0009	0.0008
S2 (strong)	53.5% [46.6%, 60.4%]	44.0%	0.0%	−0.0035	0.0020
S3 (flight to quality)	5.5% [2.3%, 8.7%]	3.5%	1.0%	0.0001	0.0013
RSC versus CCC (all pairs switch), 1,508 test days					
S0 (no correlation change)	4.6% [2.8%, 6.4%]	7.2%	4.2%	0.0004	0.0018
S1 (moderate)	17.2% [13.9%, 20.5%]	9.8%	0.4%	−0.0022	0.0032
S2 (strong)	25.4% [21.6%, 29.2%]	17.0%	0.0%	−0.0069	0.0069
S3 (flight to quality)	2.6% [1.2%, 4.0%]	6.6%	5.2%	0.0010	0.0036
Static D versus B, 1,508 test days (size under GARCH margins)					
S0 (no correlation change)	9.6% [7.0%, 12.2%]	14.8%	8.0%	0.0002	0.0024

Note: Tail level $\alpha = 5\%$, Gaussian chain, simulated data from the stylised GARCH regime-switching process (Section 5.6); S0 is the null and its row is the test size. One-sided: reject when $t < -1.6449$ (the pre-declared rule). Two-sided: $|t| > 1.96$. Upper tail: $t > +1.96$, the rate at which the richer model is declared significantly worse. The interval is a 95% binomial interval on the replications. Mean $\bar{d}$ and mean SE are averages over replications. Sources: `power_1508_summary.csv`, `power_2520_twosided.csv`.

the other way. The two-sided rates are 14.2% and 24.2% in S1 and S2. At 2,520 test days the corresponding one-sided power was 27.0% and 53.5%. Shortening the sample lowers power substantially. The pre-declared low-power rule, which would make the first sentence of the hold-out chapter say that the test cannot be expected to detect any simulated shift, applies if S2 power is below 30%; at 35.2% it does not apply. But a moderate shift would be detected about one time in five, and I report that in the hold-out table.

Simulated power does not transfer one-for-one to the real data. The mean standard error of the primary difference in the simulation depends on the scenario: it is 0.0006 in S0, 0.0010 in S1 and 0.0026 in S2, because a larger shift in correlation makes the loss difference more variable. The real-data standard errors above, 0.0010 to 0.0015, lie between the S0 and S2 values. The power of 35.2% in S2 therefore reflects both a mean difference of −0.0034 and a standard error of 0.0026, and it is not the power that the same difference would have at the

Table 6.7: SHARE OF DAYS WITH A POSITIVE 252-DAY STOCK–BOND CORRELATION, BY ERA

Era	Share of days
1987–1999	90.6%
2000–2009	23.7%
2010–2019	0.0%

Note: Mutual-fund proxies for stocks and long Treasuries (VFINX, VUSTX), not ETFs. Source: `output/explore/longhistory_summary.json`.

standard error of the real data. The minimum detectable difference at the real standard error, about $2.4865 \times 0.0014 = 0.0035$ for the one-sided test (0.0031 at the standard error observed on the hold-out, Chapter 7), is the more transferable summary. The simulated numbers describe the simulated process and not markets, and the choice of the primary test over the all-pairs version, made from the earlier 2,520-day run, is not changed by the 1,508-day result.

6.7 Why the motivating regime is absent from the development period

The question is motivated by the stock–bond correlation turning positive in 2022. Table 6.7 shows with fund proxies that the 252-day correlation was positive on 90.6% of days in 1987–1999, on 23.7% in 2000–2009 and on 0.0% in 2010–2019. No ETF development sample contains the regime the thesis is about, so the development data cannot say whether a correlation-switching model detects it. Only the hold-out can.

7 RESULTS: THE HOLD-OUT TEST

This chapter reports the single pre-specified test on the hold-out, 2 January 2020 to 31 December 2025 (1,508 forecast days), with the sentence that Section 5.4 fixed for the outcome that occurred. Secondary results follow, labelled as such, and then the calibration of the two primary models, which is the result most relevant to a practitioner.

7.1 The record of the run

The forecasts were produced once by `scripts/confirm_00_run.py` from the code fixed by the tag `spec-freeze-v1`. The runner printed 1,508 forecast days from 2020-01-02 to 2025-12-31.

Before the run the file hashes in the preregistration matched the files, the tests passed, and no earlier hold-out output existed. The supervisor had been sent the hypothesis, the sidedness and the exact SPY–IEF specification, and had replied that the test looked appropriate.

The scorer then refused to score. Its first check required the commit recorded in the forecast file to equal the tagged commit, and the recorded commit was two commits later: they changed an ignore file and the execution log, and none of the forecast code. No number from the run had been computed or seen. I replaced the check with one that requires the recorded commit to descend from the tagged commit and the forecast code to be identical in both, wrote the change and its reason into the preregistration (its section 8), and tagged it `spec-freeze-v1.1`. The forecasts were not produced again. I state this because the original rule was part of the specification, and the amendment, although it cannot depend on a result that did not yet exist, is a departure from it. It also goes beyond the letter of the preregistration, which allowed a repair only after a crash before output was written, whereas here the output existed and the scorer refused it. The new check covers `src/`, the runner and the requirements file; it does not cover configuration files, the other scripts or the Python environment. That the amendment was made before any figure was seen rests on my record and on the commit times, not on an independent witness.

The run itself was clean. None of the 144 covariance matrices needed a positive-semidefinite repair, no GARCH fit fell back to a simpler model, and none of the 24 DCC refits failed.

7.2 The pre-specified test

Table 7.1 gives the test of Section 5.3: letting the SPY–IEF correlation switch with the regime (RSC_P) against constant correlation (CCC), at tail level 5% with a Gaussian chain. The mean loss difference is -0.0013, so the switching model had the lower loss on average, with a HAC standard error of 0.0013 and $t = -1.00$. The pre-declared one-sided p-value is 0.158, and the test did not reject at the 5% level (the critical value is -1.6449). The two-sided 95% interval is from -0.0037 to $+0.0012$. The upper tail did not come near $+1.96$, so the statement that the

Table 7.1: THE PRE-SPECIFIED HOLD-OUT TEST: SWITCHING SPY–IEF CORRELATION AGAINST CONSTANT CORRELATION

Quantity	Value
Forecast days n	1,508
Mean difference $\bar{d}$ (RSC_P_N minus CCC_S_N)	−0.0013
HAC standard error (7 lags)	0.0013
t statistic	−1.00
One-sided p-value (pre-declared test)	0.158
Two-sided p-value	0.316
95% interval, two-sided	[−0.0037, 0.0012]
One-sided 95% lower bound $\bar{d} - 1.6449\,\mathrm{SE}$	−0.0033
Smallest effect of interest (SESOI)	−0.0020
MDE, one-sided 5% test, 80% power	0.0031
MDE, two-sided 5% test, 80% power	0.0035
Simulated power, moderate shift S1 / strong shift S2	18.2% / 35.2%
Outcome	O2

Note: Forecast days 2 January 2020 to 31 December 2025, tail level $\alpha = 5\%$, Gaussian chain, pooled mean. Negative $\bar{d}$ favours switching the SPY–IEF correlation. The interval is $\bar{d} \pm 1.96\,\mathrm{SE}$. Power is the simulated one-sided rejection rate at 1,508 days (Table 6.6), not a property of this estimate. Source: `output/confirm/holdout/primary.json`.

switching model is significantly worse does not apply either.

This is outcome O2, because the one-sided lower bound, −0.0033, is below the smallest effect of interest (−0.0020), so O3 is not reached. The frozen sentence is:

> The hold-out data neither establish an improvement nor exclude one. The estimate is −0.0013 (95% CI [−0.0037, +0.0012]); an improvement of at least 0.0031 (the minimum detectable difference for the one-sided 5% test, 80% power) would have been detected with 80% probability, and smaller improvements, including none, are consistent with the data.

Three further facts belong with that sentence. First, the estimate is 0.4 times the minimum detectable difference, so the design could not detect an improvement of this size with any useful probability. The standard error, 0.0013, lies within the range of 0.0010 to 0.0015 that the development data had suggested (Section 6.5). Second, the simulated power of this test at 1,508 days was 18.2% for the moderate shift and 35.2% for the strong shift (Table 6.6); these rates describe the simulated process and depend on each scenario's own standard error (0.0026 for the

Table 7.2: SECONDARY COMPARISONS ON THE HOLD-OUT (UNADJUSTED; NOT CONFIRMATORY)

Comparison	n	$\bar{d}$	SE	95% interval	t	p (2-sided)	MDE
Ladder steps (all pairs switch where stated)							
B versus A	1,508	−0.0952	0.0879	[−0.2675, 0.0771]	−1.08	0.279	0.2463
D versus B	1,508	0.0155	0.0117	[−0.0074, 0.0385]	1.32	0.186	0.0328
CCC versus B	1,508	−0.1107	0.0820	[−0.2715, 0.0501]	−1.35	0.177	0.2298
RSC versus D	1,508	−0.1299	0.0919	[−0.3099, 0.0502]	−1.41	0.157	0.2574
RSC versus CCC	1,508	−0.0037	0.0033	[−0.0101, 0.0028]	−1.11	0.265	0.0092
DCC versus RSC	1,508	−0.0060	0.0148	[−0.0351, 0.0230]	−0.41	0.683	0.0415
Primary pair at other settings							
α 2.5%, Gaussian chain	1,508	−0.0026	0.0022	[−0.0070, 0.0017]	−1.19	0.233	0.0062
α 5%, Student-t chain	1,508	−0.0012	0.0012	[−0.0035, 0.0012]	−0.96	0.339	0.0034
α 2.5%, Student-t chain	1,508	−0.0023	0.0020	[−0.0062, 0.0016]	−1.16	0.245	0.0055
Calendar year 2022 (interval only, not a test)							
RSC_P_N versus CCC_S_N, 2022	251	−0.0009	0.0041	[−0.0089, 0.0070]	–	–	–

Note: Mean FZ0 difference of the richer model minus the baseline; negative favours the richer model. Panel 1 is the ladder of Chapter 4 on the hold-out, with tail level 5%, Gaussian chain, pooled mean. The interval is $\bar{d} \pm 1.96\,\mathrm{SE}$ and the MDE is $2.8016\,\mathrm{SE}$ (two-sided 5%, 80% power). None of these p-values is adjusted and none is cited as confirmation. Source: `output/confirm/holdout/secondary_ladder.csv`, `secondary_alpha_and_t_chain.csv`, `secondary_focus_year.csv`.

strong shift, twice the observed one). At the standard error that I observed, a true improvement equal to the smallest effect of interest, 0.0020, would have been detected with a probability of about 48%; this figure was computed after the result was known, for that stated effect and not at the estimate. Third, the interval contains zero and also improvements well above the smallest effect of interest, up to −0.0037, and a degradation up to +0.0012. I do not claim equivalence with constant correlation, and I do not claim the improvement either. This is one test on one six-year sample, containing a single stock–bond reversal, and the specification was chosen with the development data in view.

7.3 Secondary comparisons

The comparisons below are the secondary analyses declared in Section 5.9. They are unadjusted, they are not confirmatory, and none of them changes the outcome above.

The ladder (the first panel of Table 7.2) can be read against the development results of Chapter 6. Letting all correlations switch (RSC against CCC) gives −0.0037 with an interval from −0.0101 to +0.0028 (two-sided $p = 0.265$, MDE 0.0092), and the smooth alternative (DCC against RSC) gives −0.0060 ($p = 0.683$). The three comparisons that involve the pooled-normal

and static models (B against A, CCC against B, RSC against D) have estimates between -0.095 and -0.130, with standard errors of 0.082 to 0.092 and MDEs of 0.23 to 0.26; their intervals contain zero, and their standard errors are more than twenty times those of the primary pair, so the hold-out says very little about them. The static step D against B has a positive estimate of $+0.0155$ ($p = 0.186$); the same oversizing caution as in Chapter 6 applies to it.

Two variants of the primary comparison change only the tail level or the chain. At $\alpha = 2.5\%$ the estimate is -0.0026 ($p = 0.233$), and with a Student-t chain at $\alpha = 5\%$ it is -0.0012 ($p = 0.339$); the estimates have the same sign as the primary one, and all intervals include zero. For the calendar year 2022 alone, the year of the stock–bond reversal, the estimate is -0.0009 over 251 days, with an interval from -0.0089 to $+0.0070$. It is reported as an interval only: a single year is too short to test, and the interval is wide enough to include a large gain and a large loss.

The Giacomini–White conditional test asks whether the loss difference depends on the lagged 63-day SPY–IEF correlation. It gives a Wald statistic of 1.40 on two degrees of freedom ($p = 0.498$). In the data, 667 of 1,508 days (44.2%) had a positive lagged correlation; the mean loss difference was -0.0004 on those days and -0.0019 on the others. These two means are descriptive and untested, and their direction is opposite to the mechanism that motivated the question. The test of superior predictive ability against GARCH-t, with the candidates B, D, RSC and RSC_P, does not reject (consistent $p = 0.477$).

7.4 Post hoc checks of the primary result

These checks were chosen after the result was known. They are descriptive, they change no outcome, and they were computed from the stored forecasts without refitting anything (`scripts/thesis_posthoc.py` and `scripts/thesis_baselines.py`).

Bandwidth. The test used a HAC lag of 7. With no lag the standard error is 0.0012 ($t = -1.04$), with 21 lags it is 0.0014 ($t = -0.90$) and with 42 lags it is 0.0015 ($t = -0.84$), so no reasonable lag turns the result into a rejection.

Where the difference arises. The sum of the daily loss differences over all 1,508 days is −1.91. By calendar year it is −0.10 in 2020, −1.56 in 2021 and −0.24 in 2022, and small afterwards. Within 2020, the days from 15 February to 30 April, which include the crash, sum to +1.05, so the switching model did worse there; excluding that window, the mean difference is −0.0020. The average is therefore the net of a few large and opposite-signed episodes, with 2021 contributing most of the favourable part and not 2022, the year of the reversal. The loss difference is also far more variable in 2020 than in 2023. A mean of such a series is a summary of one sample's history, and the normal reference for its t statistic is only as reliable as the weak-dependence assumption behind it.

Baselines. Three further forecasts of the portfolio return, scored with the same loss on the same days, did not differ detectably from constant correlation. The mean FZ0 difference against CCC (positive means worse than CCC) was +0.0317 for GARCH-t (HAC standard error 0.0429), +0.1308 for historical simulation (0.1278) and +0.0063 for the RiskMetrics EWMA (0.0440), and every interval contains zero. Their 5% breach rates were 6.6%, 5.8% and 5.2%, against 6.8% and 6.9% for CCC and RSC_P; the Kupiec test rejects correct coverage for GARCH-t ($p = 0.008$) but not for historical simulation ($p = 0.146$) or the EWMA ($p = 0.673$). A one-parameter EWMA therefore had a loss that cannot be told apart from that of the GARCH-based models and a breach rate close to nominal. This compares three further forecasts on one sample with large standard errors, and it was chosen after the result was known.

7.5 Calibration

Table 7.3 describes whether the forecasts of the two primary models had the right coverage. It is a description, not part of the decision rule.

Both models were breached more often than intended. At the 5% level the VaR was breached on 6.9% of days by RSC_P and 6.8% by CCC (104 and 103 days, against 75 expected), and the Kupiec test rejects correct coverage for both ($p = 0.001$ and 0.002). The independence tests do not reject ($p = 0.480$ and 0.444), so the breaches were not detectably clustered. At 2.5%

Table 7.3: COVERAGE AND EXPECTED-SHORTFALL CALIBRATION OF THE TWO PRIMARY MODELS ON THE HOLD-OUT

α	Model	Hits	Hit rate	Kupiec p	Indep. p	Cond. cov. p	Z_2	Z_2 p	ES ratio
5.0%	RSC_P_N	104	6.9%	0.001	0.480	0.005	0.616	<0.001	1.17
5.0%	CCC_S_N	103	6.8%	0.002	0.444	0.006	0.606	<0.001	1.18
2.5%	RSC_P_N	66	4.4%	<0.001	0.947	<0.001	1.083	<0.001	1.19
2.5%	CCC_S_N	65	4.3%	<0.001	0.903	<0.001	1.064	<0.001	1.20

Note: 1,508 forecast days. A hit is a day with $y_t \leq \mathrm{VaR}_t$. Kupiec: unconditional coverage; Indep.: Christoffersen independence; Cond. cov.: their sum. Z_2 is positive when risk is underestimated (my sign, opposite to Acerbi and Szekely); its p-value is the simulated share of Z_2 values at or above the observed one. ES ratio: mean of y_t/ES_t over hit days, above one when exceedances are larger than the forecast shortfall. Descriptive, not part of the decision. Source: `output/confirm/holdout/calibration.csv`.

the breach rates were 4.4% and 4.3%, and the Kupiec p-values are below 0.001. For expected shortfall, Z_2 is positive for both models (0.62 and 0.61) with a simulated p-value of 0.0002, the smallest the 5,000 simulated draws can give. Z_2 combines the frequency of breaches with their size, so much of it reflects the excess of breaches over the 5% expected; the size alone is shown by the average return on breach days, about 1.17 to 1.18 times the forecast shortfall, which I report without a separate test. In this hold-out, therefore, breaches were too frequent and breach-day losses exceeded the forecast shortfall, to a similar degree in both models. The independence test is first-order and has little power with about a hundred breaches, so the absence of detectable clustering is weak evidence. This contrasts with the development data, where the static models overstated risk and the GARCH-margin models were close to nominal (Table 6.3). Both models were miscalibrated by about the same amount, so the small loss difference between them says nothing about whether either is adequate for risk management.

8 DISCUSSION AND LIMITATIONS

This chapter says what the evidence supports, what it leaves open, and what limits it.

8.1 What the evidence shows

The pre-specified test did not reject (outcome O2, Chapter 7). The data are consistent with a gain of the size that the development data had suggested and with a zero difference. The result is not against regime-switching correlation and not for it.

The development data add three descriptive findings, none of them a test with a controlled error rate: regime-specific volatility lowered the loss against a pooled normal in three of four ETF samples; a GARCH-margin model class did better than static regime covariance in all four, established after adjustment in one; and the correlation steps were small relative to their minimum detectable differences. I examined many specifications to reach these, so I treat them as the reason the thesis has its shape and not as confirmed. The decomposition is the part I would defend most: for each step it gives an estimate, an interval and what the design could detect.

The calibration result is the one most relevant to a practitioner, although it covers only the two primary models and its explanation is untested. Both breached the 5% VaR on about 6.9% of hold-out days and, on breach days, losses exceeded the forecast expected shortfall. Which of the two is better is secondary to the fact that both fall short, and the shared shortfall is one reason their loss difference is small. In the development data the GARCH-margin models were close to nominal, with breach rates of 4.3% to 5.5% in the ETF samples (Table 6.3), so the 2020–2025 shortfall is consistent with the 2020 crash and the 2022 reversal being unlike the exploration sample. I did not test that explanation. The primary chain is Gaussian, and Paolella et al. (2019) report that heavy tails, more than correlation dynamics, determine VaR accuracy. The Student-t variant of the primary comparison has the same sign as the primary estimate (Section 7.3); its calibration is not reported here. A post hoc RiskMetrics EWMA of the portfolio return breached the 5% VaR on 5.2% of days, and its loss exceeded that of CCC by 0.0063 (HAC standard error 0.0440; Section 7.4). The simplest practitioner baseline was therefore better calibrated on this sample, and its loss difference is small against its own standard error.

8.2 What the evidence does not show

Three readings are not supported. First, that switching the correlation fails to improve forecasts: the interval includes improvements above the smallest effect of interest, and the lower bound

of −0.0033 is below −0.0020. Second, that switching does help: the test did not reject, and the 2022 estimate, −0.0009 with an interval from −0.0089 to +0.0070, is too imprecise to say anything about the episode that motivated the question. Third, that the volatility regime, the correlation regime or the GARCH margins is responsible for any difference: the comparisons change several things at once (Table 4.1), and the non-nested contrasts compare model classes.

8.3 Why a volatility-regime signal and a long window may be slow

The regime is inferred from SPY returns alone, so a two-state volatility model separates calm from stressed days. Campbell et al. (2020) tie the earlier switch to a change in the cyclicality of inflation, a macroeconomic state that changes over years, while Baele et al. (2010) find that macroeconomic fundamentals explain little of the correlation (Section 2.4). If the state is macroeconomic and slow, it need not coincide with high equity volatility; if it is market-based, an equity-volatility signal may be closer to the right one. The result does not distinguish the two. The stress-day correlation is also learned from the window, and before 2022 that window held almost no positive stock–bond correlation (Table 6.7), so the model had little with which to represent a positive stress correlation however quickly it moved. Each model is estimated on the last 1,260 days and refitted every 21 days, and in 2022 the correlation changed within a window that still held several years of the old relationship. Both choices were declared as limits before the run (Section 5.10). I did not test either explanation or run a faster-window or different-signal variant on the hold-out, because that analysis was not pre-specified.

8.4 Limitations

The limits of Section 5.10 apply. Five matter most for reading the result.

- **Power.** The test could detect a difference of 0.0031, which exceeds the smallest effect of interest, 0.0020, so outcome O3 required an estimate slightly above zero (Section 5.7). The test is better read as a measurement of how precisely this sample can speak than as a test with a realistic chance of excluding a meaningful gain. A longer ETF history would

not add the regime of interest, because before 2020 the ETF data hold almost no positive stock–bond correlation (Table 6.7).

- **One test, chosen with the development data in view.** The comparison was selected from a simulation in which the SPY–IEF shift is largest by construction, after I had examined ten universes and ten regime definitions. Nothing was fitted on the hold-out, but the specification is not independent of my exploration.

- **A stylised power study.** The simulated process is an assumption, and simulated and real-data standard errors differ (Chapter 6), so the power figures describe the simulated process and need not transfer to markets.

- **Inference.** The estimand is the expected loss difference of two forecasting procedures, each with its own estimation window, in the sense of Giacomini and White (2006). The standard error does not carry the uncertainty of the HMM and GARCH fits into a claim about the population. The 2020 and 2021 episodes dominate the average (Section 7.4). The 3.6% rejection rate in the simulated null (500 replications, interval 2.0% to 5.2%) is a rough size check, and the upper-tail rate in that scenario, 8.0% at 1.645, shows that the normal reference is not exact there.

- **Vendor and scope.** All prices come from one vendor, and the portfolio is three funds with constant weights. The decision is made at the 5% tail level, and I make no claim about other portfolios, tail levels or markets.

8.5 Future work

Four extensions follow, each a new study with its own pre-specification. A regime signal built from the stock–bond correlation or from macroeconomic variables would test directly whether a slow state helps where a volatility state may not. A window that adapts to the regime, or exponentially weighted estimates, would address the slowness described above. A longer or

broader sample, such as several portfolios or international funds, would raise the power, and a power calculation like the one in Chapter 5 should come first. A bootstrap of the whole estimation pipeline would carry the uncertainty of the regime estimates into the test.

9 CONCLUSION

I asked whether letting the stock–bond correlation switch with a market regime improves one-day value-at-risk and expected-shortfall forecasts of an equal-weight portfolio of SPY, IEF and GLD. I built a ladder of models that separates the treatment of volatility, correlation and the mean under one regime signal, fixed a single confirmatory test before using 2020–2025, and ran it once.

The test could not tell a gain from none. The estimate favoured switching by 0.0013, the pre-specified test did not reject (outcome O2, Chapter 7), the sample could not have detected a gain smaller than 0.0031, and a non-significant result is not evidence that the effect is absent.

For a risk manager of a portfolio like this one, the evidence gives no reason to prefer the more complex model, and it does not rule out that the model helps. It also shows a more immediate problem: both primary models breached the 5% VaR on about 6.8% to 6.9% of hold-out days and breach-day losses exceeded the forecast expected shortfall, so neither forecast was well calibrated over 2020–2025. A simple EWMA of the portfolio return, scored post hoc, breached on 5.2% of days with a loss that cannot be told apart from theirs. On the development data, regime-specific volatility lowered the loss in three of four ETF samples, and the correlation steps were small beside their minimum detectable differences.

The limits are those of Chapter 8: one portfolio, one vendor, one reversal episode, a regime signal built from equity returns, and a power study that rests on a stylised process. Whether a slower regime signal or a larger sample would show the gain that this one could not is a question for a new, separately pre-specified study.

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APPENDIX

A SPECIFICATION, REPRODUCIBILITY AND FROZEN WORDING

A.1 Where the record is

Every methodological decision is in `docs/decisions_log.md`, including the exploration grid (10 ETF universes, 10 regime definitions, 214 headline tests) and the family tests. The confirmatory specification is `docs/preregistration_confirmatory.md`, fixed by the annotated tags `spec-freeze-v1` and `spec-freeze-v1.1`; its section 8 records the one amendment (Section 7.1). `docs/EXECUTION_LOG.md` lists each step of the work in order, with the hashes of the hold-out forecast file and of the scored tables. The hold-out forecast file has SHA-256 `9537bb5d6008028076126d0166bb2edc1f643e5a9f496cb852d7872f8485f974`; the scored tables are in `output/confirm/holdout/` with a manifest of their hashes. Tables of the development controls and sub-periods are in `output/explore/` and are not reprinted.

A.2 Rebuilding the thesis from saved results

The command `python scripts/build_all.py` rebuilds every table of Chapters 5 to 7, the forest plots and the PDF, and checks every registered number against its source file. It reads the saved result files, refits nothing and takes a few seconds. `scripts/verify_clean_clone.py` repeats the build in a clean clone of the committed repository and compares each table byte for byte, each figure pixel by pixel and the text of the PDF.

A.3 Recomputing the results

The saved results can be recomputed from the raw data file stored under `data/raw/` (SHA-256 `0ab8c7f3...150e29`) in the order below.

- The exploration grid, the ladder and the development power study: `scripts/explore_00` to `explore_20`.
- Power at the hold-out length: `scripts/explore_21_power_holdout_length.py`, 500 repli-

cations in each of four scenarios (about 19 seconds per replication), summarised by `scripts/explore_22_p`

- The confirmatory dry run and hold-out: `scripts/confirm_00_run.py pseudo` (18 seconds) and `holdout` (27 seconds), then `scripts/confirm_01_score.py`. The runner refuses to run the hold-out unless the frozen files match the tag and refuses to overwrite an existing output.

The test suite (`pytest`; 105 tests in about one minute, plus three that read the hold-out period) checks the models, the scorer and the helper scripts.

A.4 Environment

Python 3.9.6 with numpy 2.0.2, pandas 2.3.3, scipy 1.13.1, arch 7.2.0, statsmodels 0.14.6, hmmlearn 0.3.3 and matplotlib 3.9.4; the full list is in `requirements.txt`. BLAS threads are set to one so that the numbers should not depend on the core count; I did not test this on another machine.

A.5 Hidden Markov model initial values

The first fit of a run is a cold start with 100 iterations from start probabilities $(0.5, 0.5)$, a transition matrix with diagonal $(0.98, 0.95)$, both means at the window mean, and variances $0.5v$ and $2.5v$, where v is the window variance.

A.6 Frozen wording of the other outcomes

The wording fixed before the run for the outcomes that did not occur was:

O1 "The hold-out data provide statistical evidence at the one-sided 5% level that letting the SPY–IEF correlation switch with the regime lowers expected FZ0 loss relative to constant correlation (difference $\bar{d}$, 95% CI $[a, b]$). This is one test on one six-year sample containing a single stock–bond reversal; simulated power was P%."

O3 "The data are inconsistent, at the one-sided 5% level, with an improvement larger than SESOI = 0.002 FZ0 units. Improvements smaller than this, and any degradation, are not

excluded."

O4 "The hold-out data provide evidence that the switching specification has higher expected FZ0 loss ($\bar{d}$, 95% CI $[a, b]$)."